

TANANA, AK, USA

WMO: 701780

Lat: 65.175N

Lon: 152.107W

Elev: 222

StdP: 14.58

Time Zone: -9.00 (NAK)

Period: 99-19

WBAN: 26529

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -47.4	(c) -41.3	(d) -34.3	(e) 0.8	(f) -28.3	(g) -32.5	(h) 0.9	(i) -27.1	(j) 19.1	(k) 1.5	(l) 17.4	(m) 2.6	(n) 0.9	(o) 280	(p) 0.691

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.8	79.5	61.2	76.1	59.8	73.0	58.5	63.7	75.2	62.0	72.5	60.2	69.8	5.8	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
59.2	75.8	66.0	57.4	70.9	65.2	55.8	66.8	64.1	29.0	75.2	27.8	72.2	26.7	69.8	73.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation	n=5 years		n=10 years		n=20 years		n=50 years		DB	WB
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min		
(a) 16.6	(b) 14.4	(c) 12.4	(d) 10.0	(e) -49.6	(f) 84.5	(g) 5.6	(h) 2.4	(i) -53.6	(j) 86.2	(k) -56.9	(l) 87.6	(m) -60.1	(n) 88.9	(o) -64.2	(p) 90.6	
(a) 16.6	(b) 14.4	(c) 12.4	(d) 10.0	(e) -34.9	(f) 66.7	(g) 3.1	(h) 2.7	(i) -37.1	(j) 68.6	(k) -38.9	(l) 70.2	(m) -40.7	(n) 71.8	(o) -42.9	(p) 73.8	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	26.6	-8.0	1.2	7.1	27.9	46.9	58.4	60.2	53.9	43.3	26.3	4.0	-3.9		
	DBStd	27.23	19.46	14.98	12.60	11.61	8.62	5.43	5.23	5.58	7.17	9.65	13.62	15.98		
	HDD50	9352	1798	1366	1329	664	159	4	3	24	216	737	1381	1671		
	HDD65	14052	2263	1786	1794	1114	560	208	163	346	650	1201	1831	2136		
	CDD50	800	0	0	0	0	64	256	319	145	16	0	0	0		
	CDD65	25	0	0	0	0	0	9	15	1	0	0	0	0		
	CDH74	500	0	0	0	0	37	182	245	36	0	0	0	0		
	CDH80	67	0	0	0	0	4	26	35	2	0	0	0	0		
	WSAvg	4.9	4.7	5.5	5.5	5.7	5.4	4.9	4.5	4.3	4.7	5.1	4.5	4.7		
	PrecAvg	13.0	0.6	0.5	0.6	0.4	0.6	1.4	2.2	2.6	1.6	0.9	0.7	0.8		
	PrecMax	20.9	2.4	1.9	2.2	1.3	1.1	2.6	5.0	5.1	5.0	2.2	2.2	2.2		
(15) Precipitation	PrecMin	5.9	0.0	0.0	0.0	0.0	0.0	0.2	0.4	0.9	0.0	0.1	0.0	0.1		
	PrecStd	3.9	0.5	0.4	0.6	0.4	0.3	0.7	1.1	1.1	1.3	0.5	0.5	0.6		
	DB	30.7	33.6	38.8	59.4	78.8	83.4	84.0	78.6	65.9	50.1	33.6	28.4			
	MCWB	29.1	32.2	34.4	45.3	56.0	61.8	63.9	62.6	53.0	42.9	32.0	26.3			
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	25.8	29.2	33.8	53.5	72.5	79.4	79.9	73.4	62.3	44.6	30.0	24.8			
	2% MCWB	24.1	26.5	28.9	41.8	54.3	60.1	62.6	60.2	51.9	40.4	28.3	23.2			
	5% DB	21.6	25.6	29.8	48.9	67.9	75.6	76.5	68.9	58.7	41.2	26.7	20.9			
	5% MCWB	20.3	23.6	26.2	39.5	51.7	58.5	61.3	57.8	50.3	37.9	25.2	19.5			
	10% DB	17.4	21.3	25.8	44.8	63.9	72.6	73.3	65.3	55.3	38.4	22.5	17.3			
	10% MCWB	16.0	19.5	22.7	37.2	50.1	57.0	60.2	56.4	49.4	35.7	21.3	16.0			
	0.4% WB	29.7	32.2	35.3	46.1	57.5	64.6	66.4	64.0	55.2	45.5	32.4	27.2			
	0.4% MCDB	31.2	33.4	38.3	56.6	74.6	79.0	79.0	75.2	61.9	48.9	33.2	28.3			
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	24.9	27.0	29.4	42.4	55.2	62.2	64.3	61.6	53.0	40.9	28.4	23.4			
	2% MCDB	26.4	28.8	32.5	52.2	69.6	74.9	75.9	70.4	59.5	43.9	29.8	24.8			
	5% WB	21.1	23.7	26.2	40.1	53.2	60.3	62.9	59.5	51.4	38.2	25.3	19.9			
	5% MCDB	22.4	25.7	29.8	48.3	65.5	72.2	73.5	66.2	57.0	40.7	26.7	21.0			
	10% WB	17.2	19.8	22.8	37.4	51.0	58.6	61.5	57.7	50.0	35.9	21.4	16.6			
	10% MCDB	18.6	21.5	26.0	44.0	61.8	69.5	71.3	63.9	55.3	38.2	22.6	17.9			
	MDBR	14.1	16.5	22.9	21.9	23.1	22.5	19.8	18.5	16.7	11.9	13.1	13.3			
	5% DB	12.0	13.5	22.5	24.0	30.6	29.3	28.1	25.9	23.6	13.5	10.5	12.1			
(36) Mean Daily Temperature Range	5% WB	11.3	12.2	18.5	14.6	15.2	14.0	14.0	14.5	14.8	9.9	9.6	11.0			
	MCDBR	11.8	13.6	21.6	22.8	27.5	25.0	24.5	20.2	19.5	12.4	10.0	11.9			
	MCWBR	11.1	12.5	17.9	14.1	13.8	12.7	13.3	12.1	12.8	9.5	9.2	11.0			
	MCWBR	11.1	12.5	17.9	14.1	13.8	12.7	13.3	12.1	12.8	9.5	9.2	11.0			
(40) Clear-Sky Solar Irradiance	taub	0.175	0.243	0.253	0.326	0.350	0.379	0.382	0.389	0.310	0.270	N/A	0.177			
	taud	1.925	2.205	2.293	2.178	2.385	2.328	2.363	2.332	2.542	2.402	N/A	1.935			
	Ebn at Noon	110	212	268	265	272	264	259	243	248	203	N/A	29			
	Edh at Noon	7	16	24	35	32	35	33	31	20	14	N/A	2			
(44) All-Sky Solar Radiation	RadAvg	46	256	830	1464	1676	1770	1501	1118	705	278	71	11			
	RadStd	5	21	59	92	147	124	104	113	92	36	10	1			

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+2.27	N/A	N/A	N/A	N/A	N/A	N/A	-884	+200	+23
(47) Regional (5 neighbors)		+1.87	N/A	N/A	N/A	N/A	N/A	-576	-700	N/A	N/A

Nomenclature: See separate page